

CS & IT ENGINEERING



Digital Logic

Optimisation

Lecture No. 1



By- Aditya sir

Recap of Previous Lecture



Topic



Universal Gates

NAND, NOR

Topics to be Covered

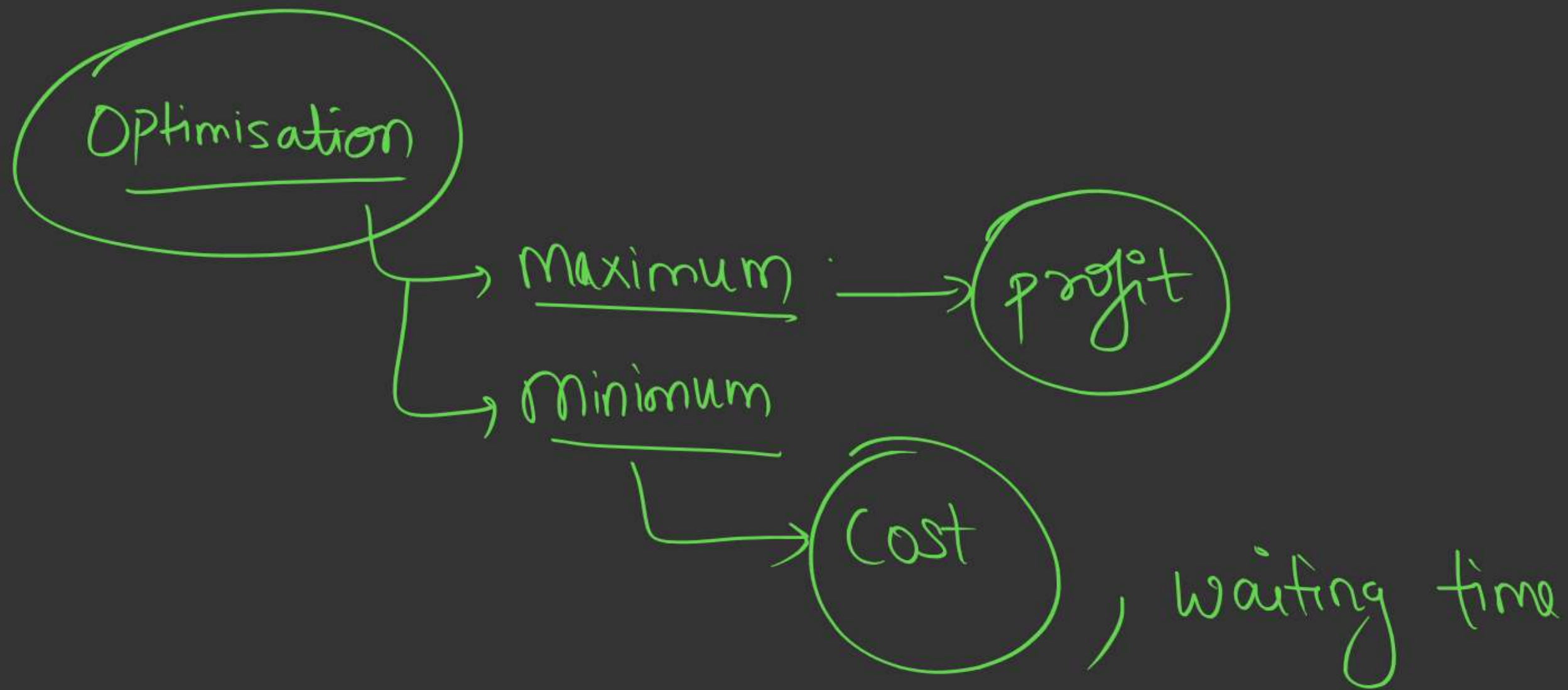


Topic



Min. no of universal Gats

→ Optimization



Digital Logic

$$y = \underline{\underline{AB + CD}}$$

[Digital Circuit]

Optimization

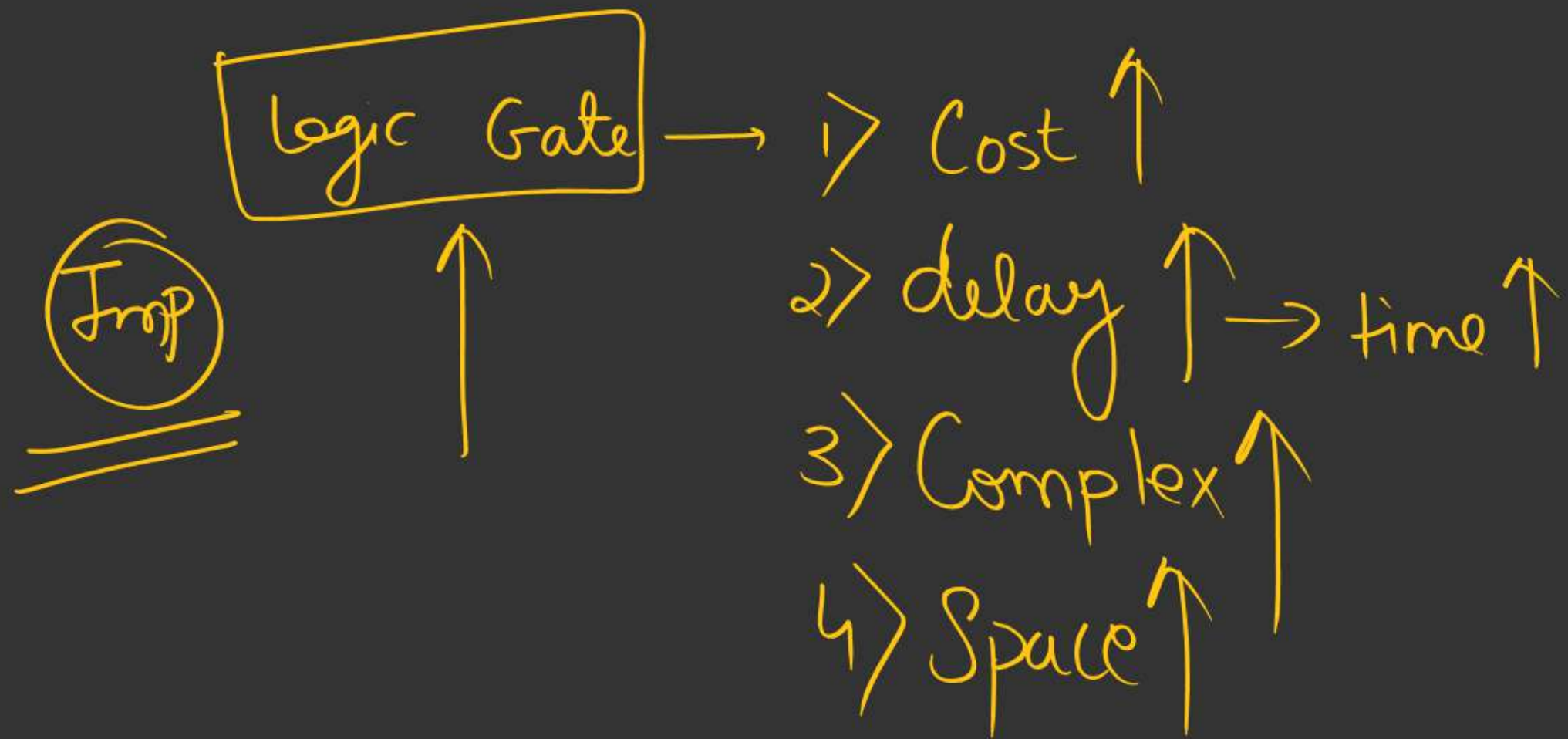
→ minimization

↓
(no. of ^{Logic} Gates)

↓
Gates

[AND, OR, NOT, XOR,
XNOR,
NAND, NOR]

why lesser Gates?

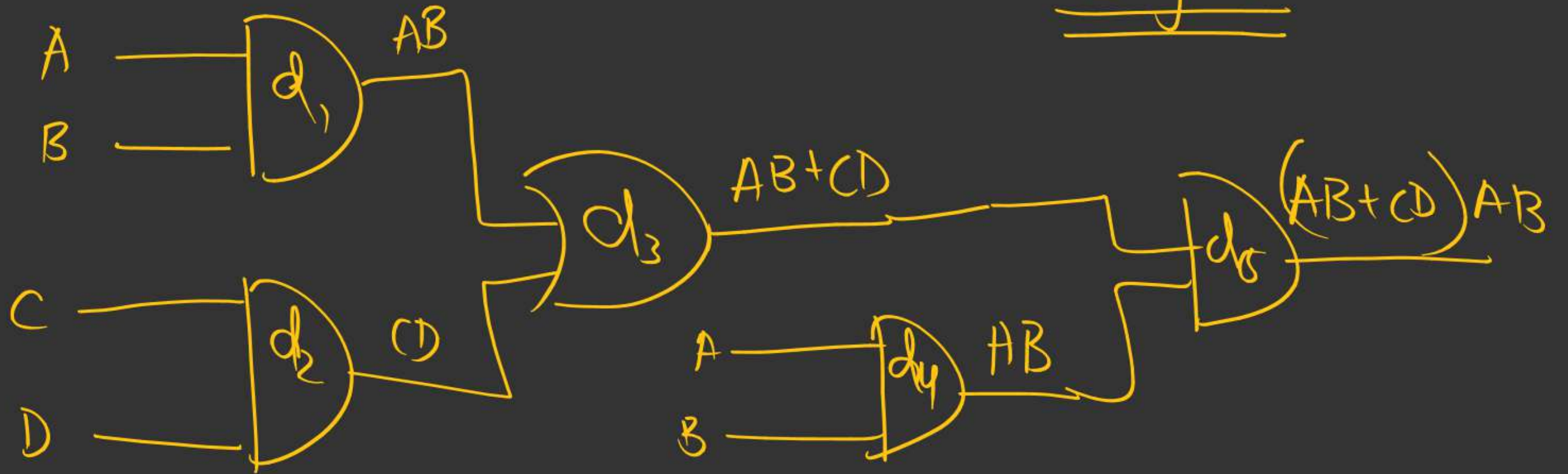


Optimization

Cost ↓
delay ↓
time ↓
Complex ↓
Space ↓

$$y = (AB + CD) \cdot AB$$

5 gates



16

A	B	C	D

$$y = \underline{(AB + CD)AB} = \underline{AB} \checkmark$$

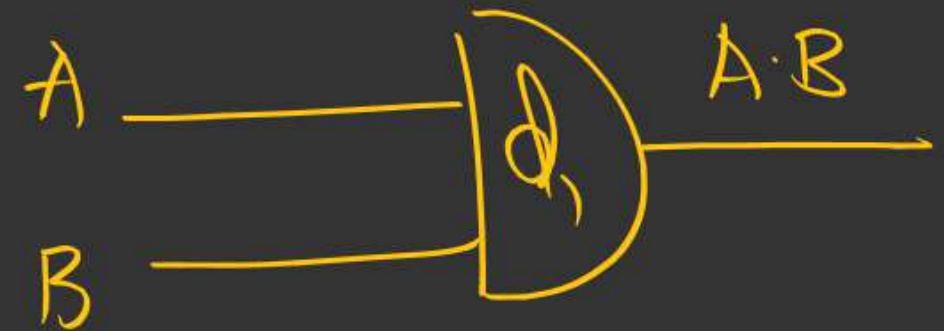
$$= AB \cdot AB + AB \cdot CD$$

$$= AB + AB \cdot CD$$

$$= AB[1 + CD]$$

$$= \textcircled{AB}$$

1 gate

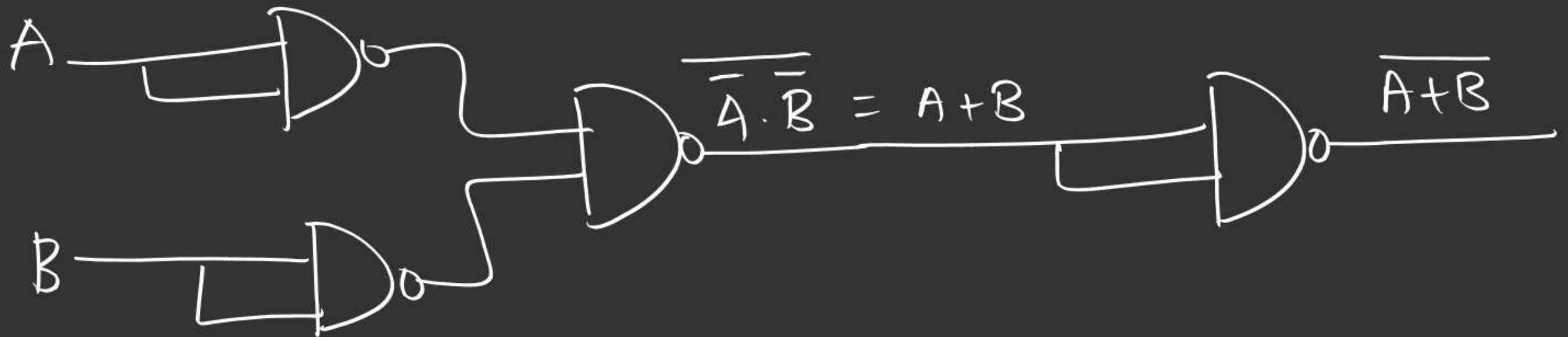


NOR using NAND

OR + NOT

(3) + (1) = 4 NAND

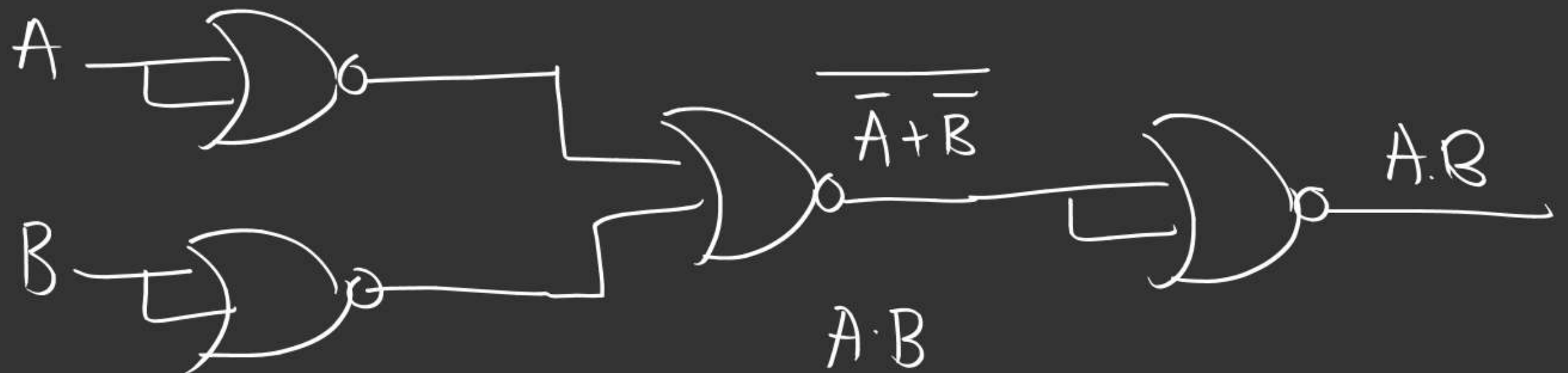
(4)



NAND using NOR

NAND $\rightarrow$ AND + NOT

$$(3) + (1) = \underline{4}$$



Gate	<u>NAND</u>	<u>NOR</u>
NOT	<u>1</u>	<u>1</u>
<u>AND</u>	2	3
<u>OR</u>	3	2
XOR	4	5
XNOR	5	4
NAND	1	4
NOR	4	1

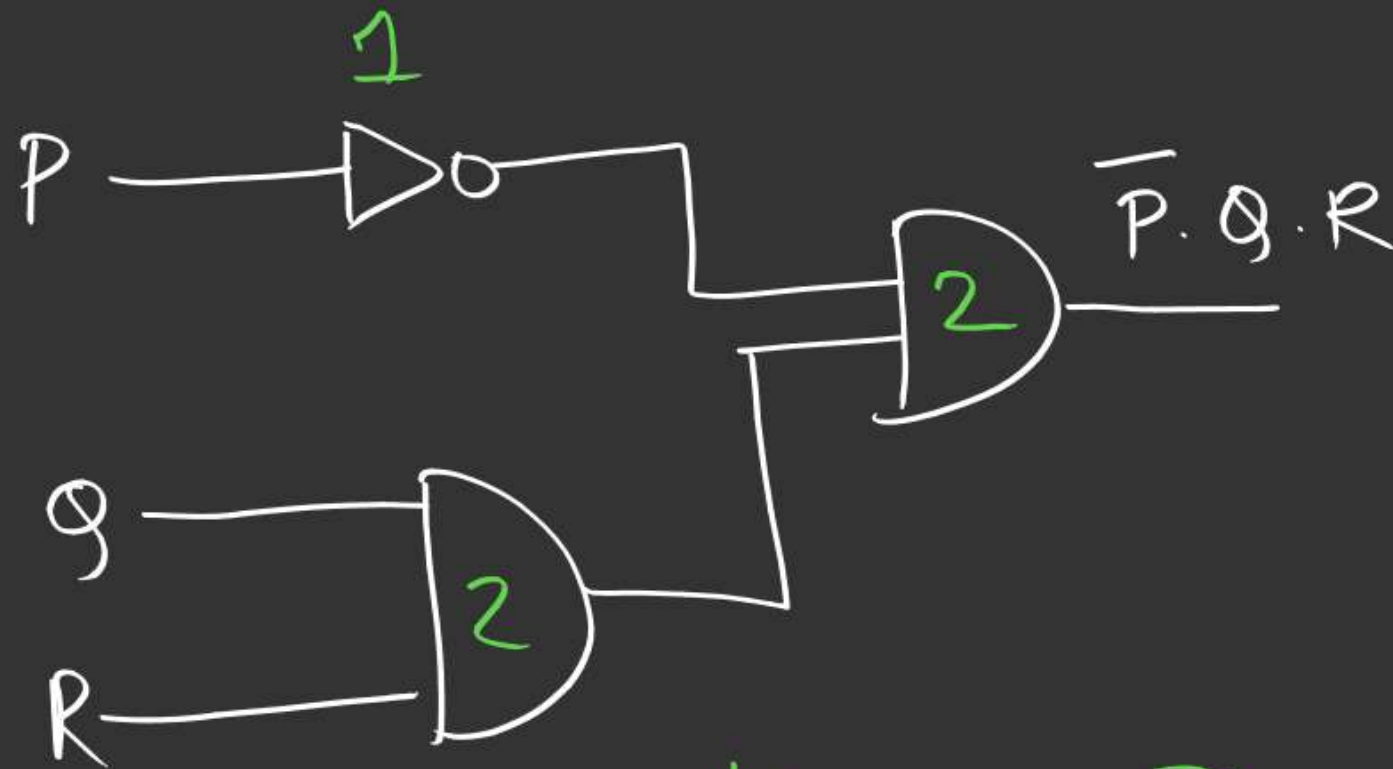
(Q.1) Find the min. no of 2 i/p NAND gate reqd to implement the given logic:

(32%)

$$f(P, Q, R) = \bar{P} \cdot Q \cdot R$$

- A) 3
- B) 4
- ☒ C) 5
- D) None

NOT $\rightarrow$ 1
AND $\rightarrow$ 2
OR $\rightarrow$ 3



$$1 + 2 + 2 = 5 \checkmark$$

AOI

AND
OR

Inverted (NOT)

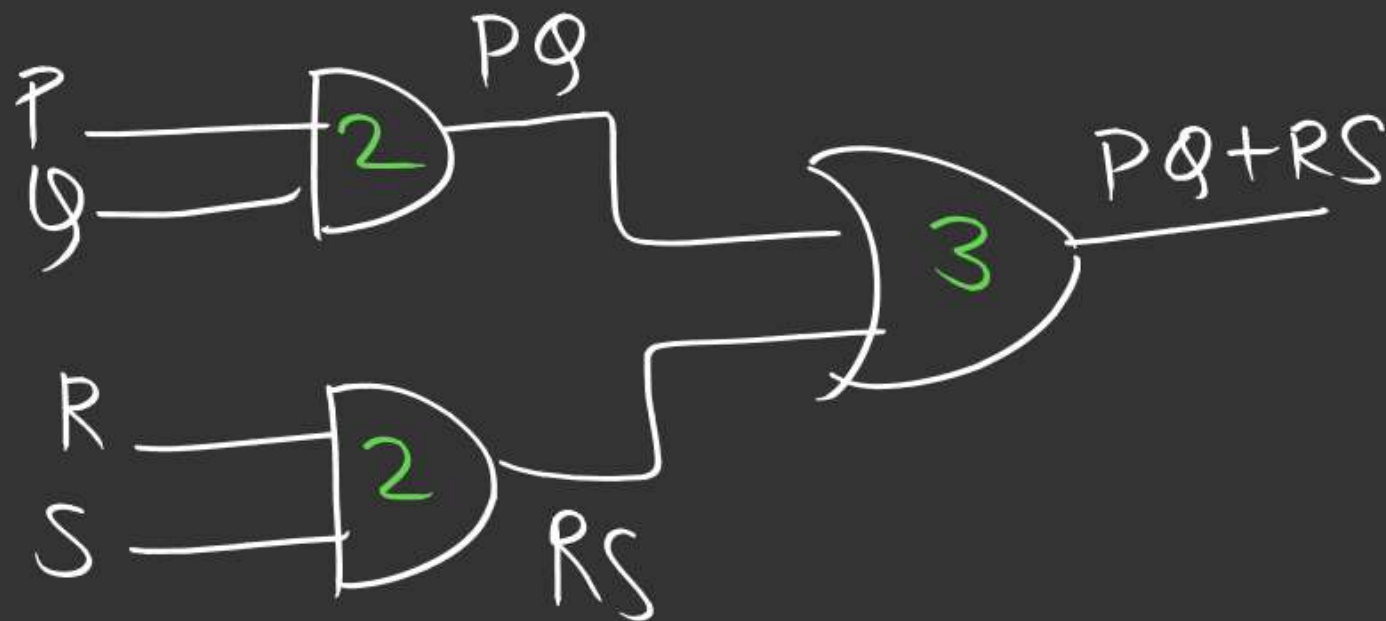
(Q.2) Min. no of 2 i/p NAND gates
to implement

$$F(P, Q, R, S) = PQ + RS$$

$$y = PQ + RS$$

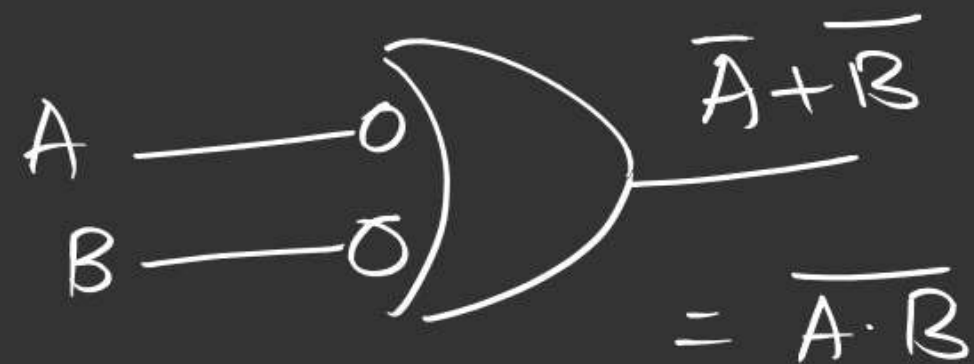
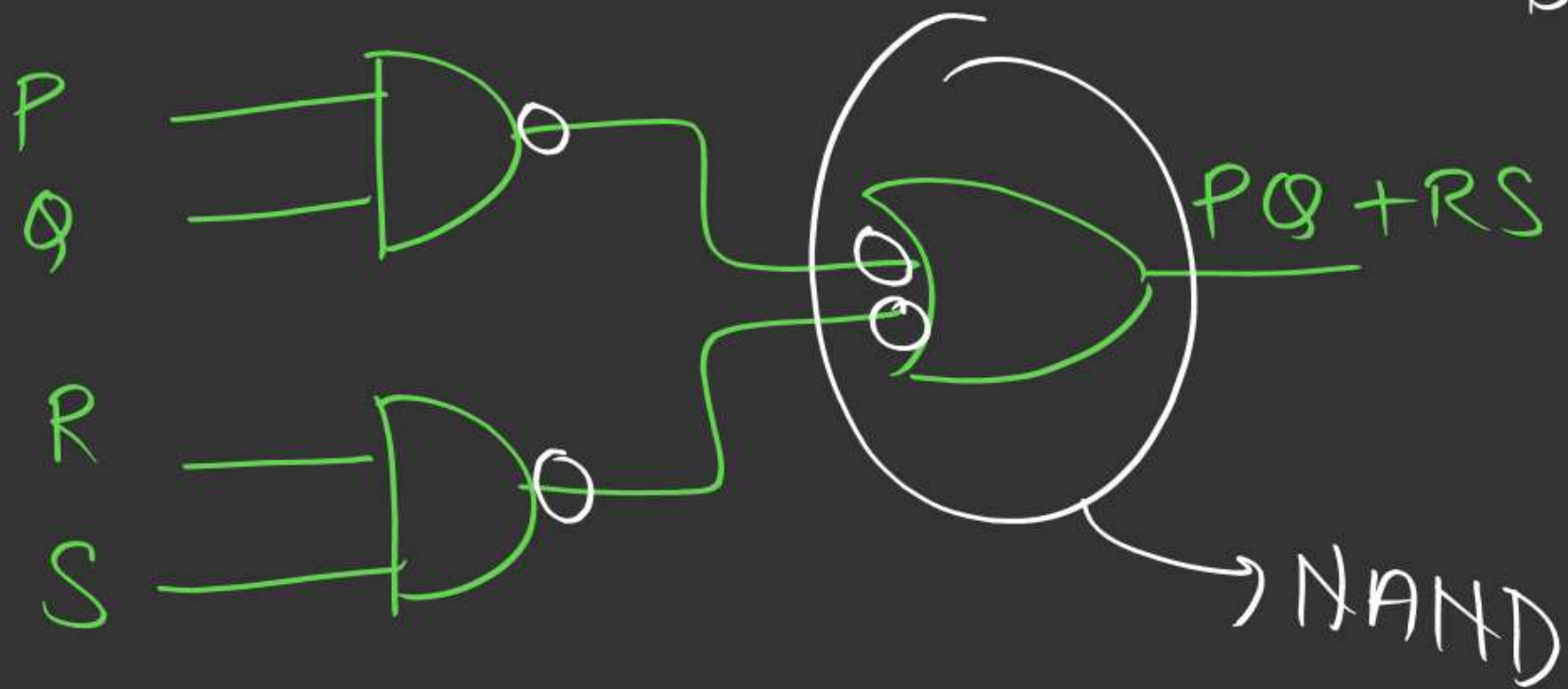
(25%)

- A) 1
~~B) 7~~ X
C) 3
D) 5



$$2 + 2 + 3 = 7 \text{ NAND}$$

(not min)



3 NAND

Case 1: all i/p's in AND format

eg: $y = P \cdot \overline{Q} \cdot \overline{R} \cdot S \cdot T$

1) NAND gates = $(2n - 2) + K$

2) NOR gates = $(3n - 3) - K$

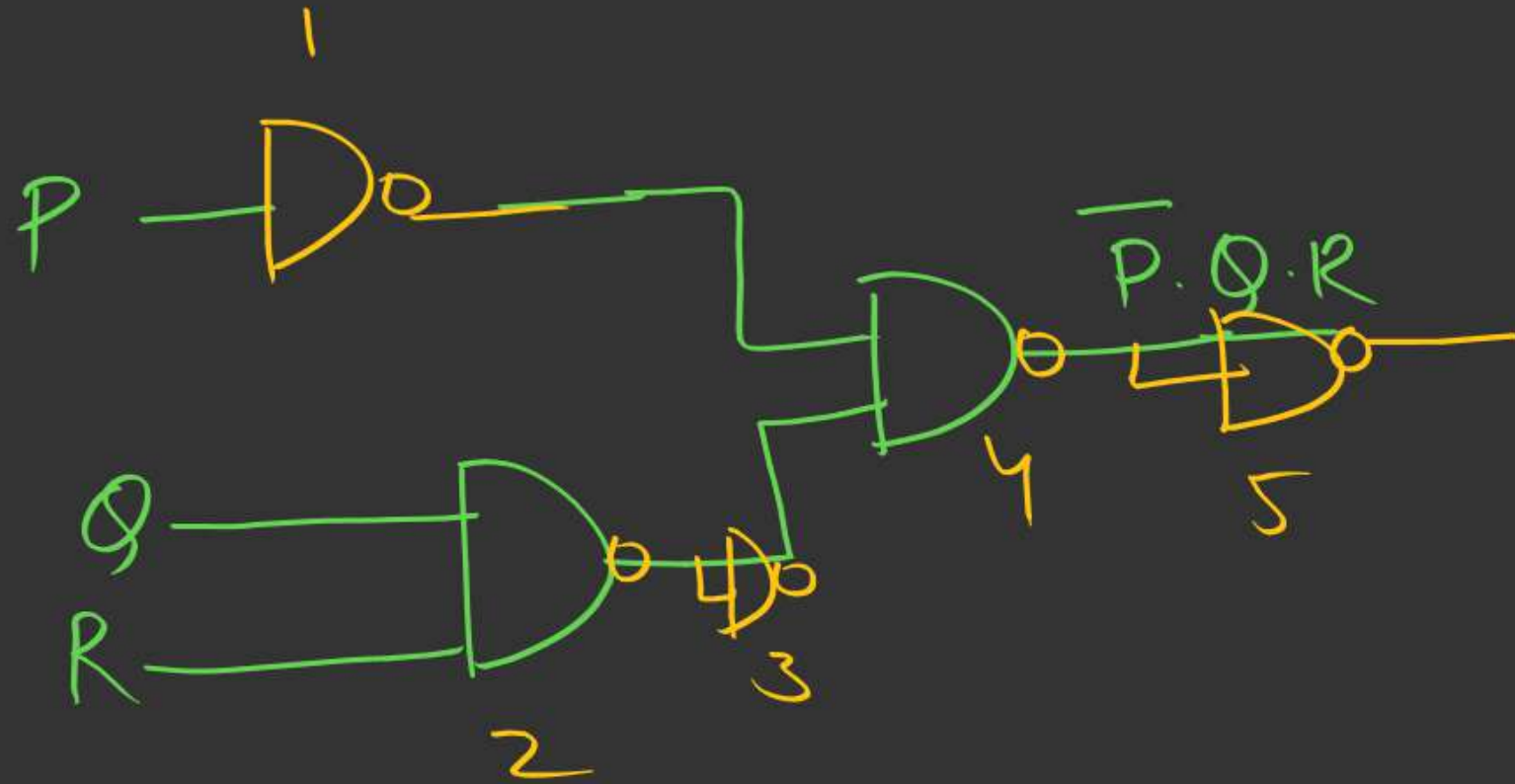
, n = total no. of variables
 K = total no. of
complimented variables
default $\rightarrow$ 2 i/p gates
no i/p in lamp form

eg1 $y = \overline{P} \cdot Q \cdot R$

$n = 3$

$k = 1$

NAND gates $= (2n - 2) + k$
 $= (2 \times 3 - 2) + 1$
 $= (6 - 2) + 1 = \textcircled{5}$



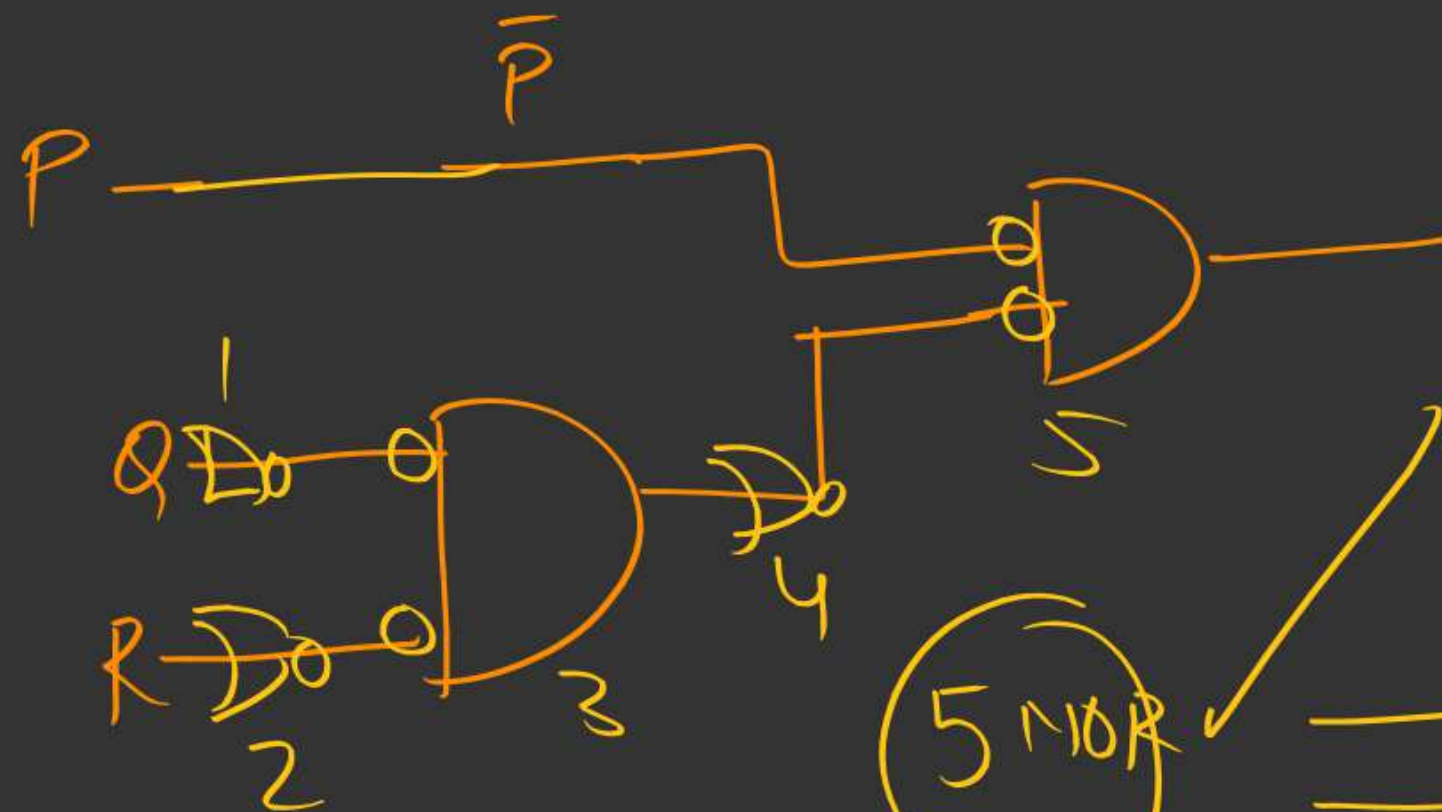
$$y = \bar{P} \cdot Q \cdot R$$

NOR

$$(3n-3) - k$$

$$n=3, k=1$$

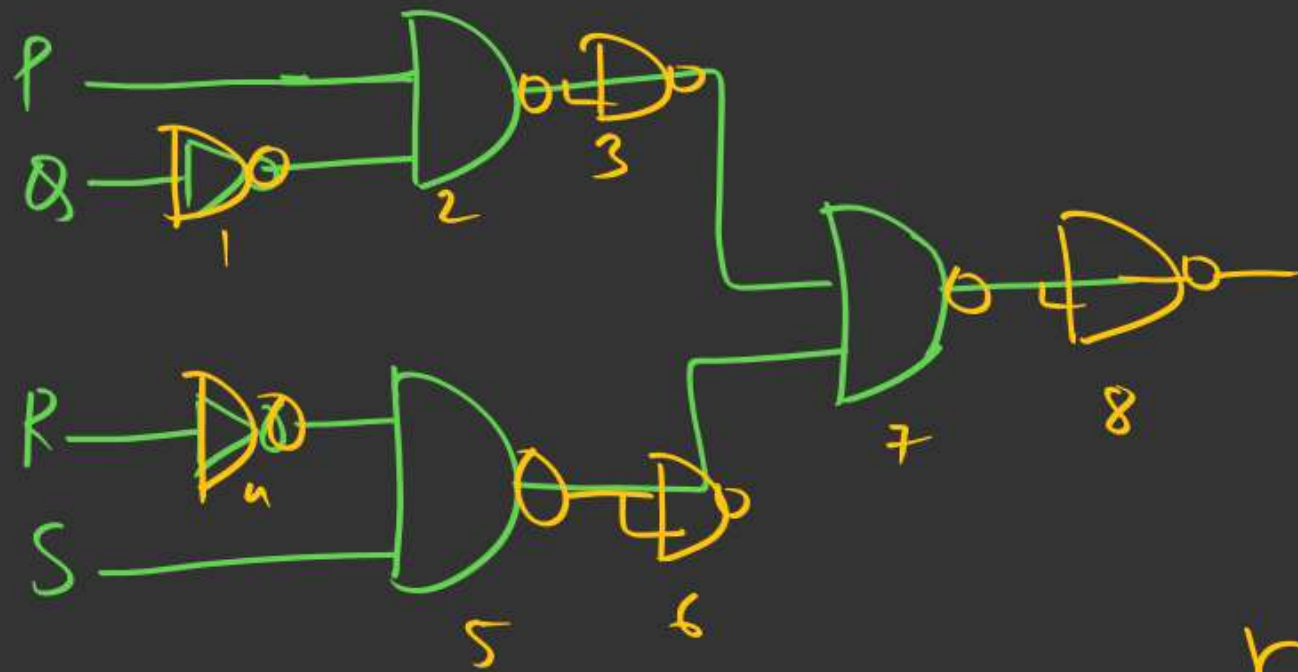
$$3 \times 3 - 3 - 1 = 9 - 4 = \textcircled{5}$$



$\textcircled{5 \text{ NOR}}$



eg 2:- $y = (P \cdot \overline{Q}) \cdot (\overline{R} \cdot S)$



8 NAND

$$n=4$$

$$K=2$$

formula:

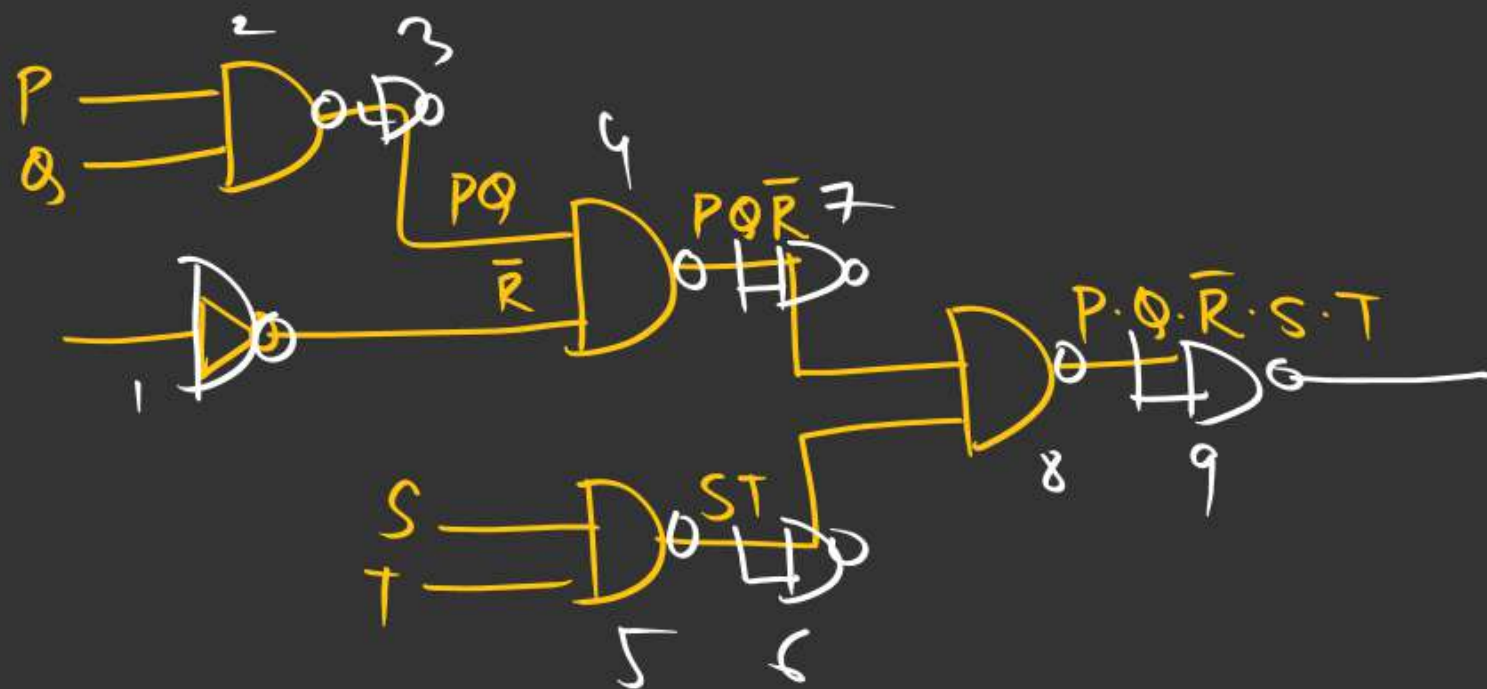
$$(2n-2) + K$$

$$= (2 \times 4 - 2) + 2$$

$$= (8 - 2) + 2 = \textcircled{8} \checkmark$$

eg 3) $y = P \cdot Q \cdot \bar{R} \cdot S \cdot T$

Default
 $\left[\begin{array}{l} 2 \text{ i/p gates} \\ \text{no comp i/p} \end{array} \right.$



9 NAND

$n = 5, k = 1$

$$(2n - 2) + k = (2 \times 5 - 2) + 1$$

$$= 8 + 1 = \boxed{9}$$

eg 4)

$$y = \overline{P} \overline{Q} R \overline{S}$$



min NAND, NOR

$$n=4, k=3$$

NAND,

$$\begin{aligned} & (2n-2) + k \\ &= (2 \times 4 - 2) + 3 \\ &= 6 + 3 = \boxed{9} \end{aligned}$$

NOR

$$\begin{aligned} & (3n-3) - k \\ &= 3 \times 4 - 3 - 3 \\ &= 12 - 6 \\ &= \textcircled{6} \end{aligned}$$

H.W

NOR

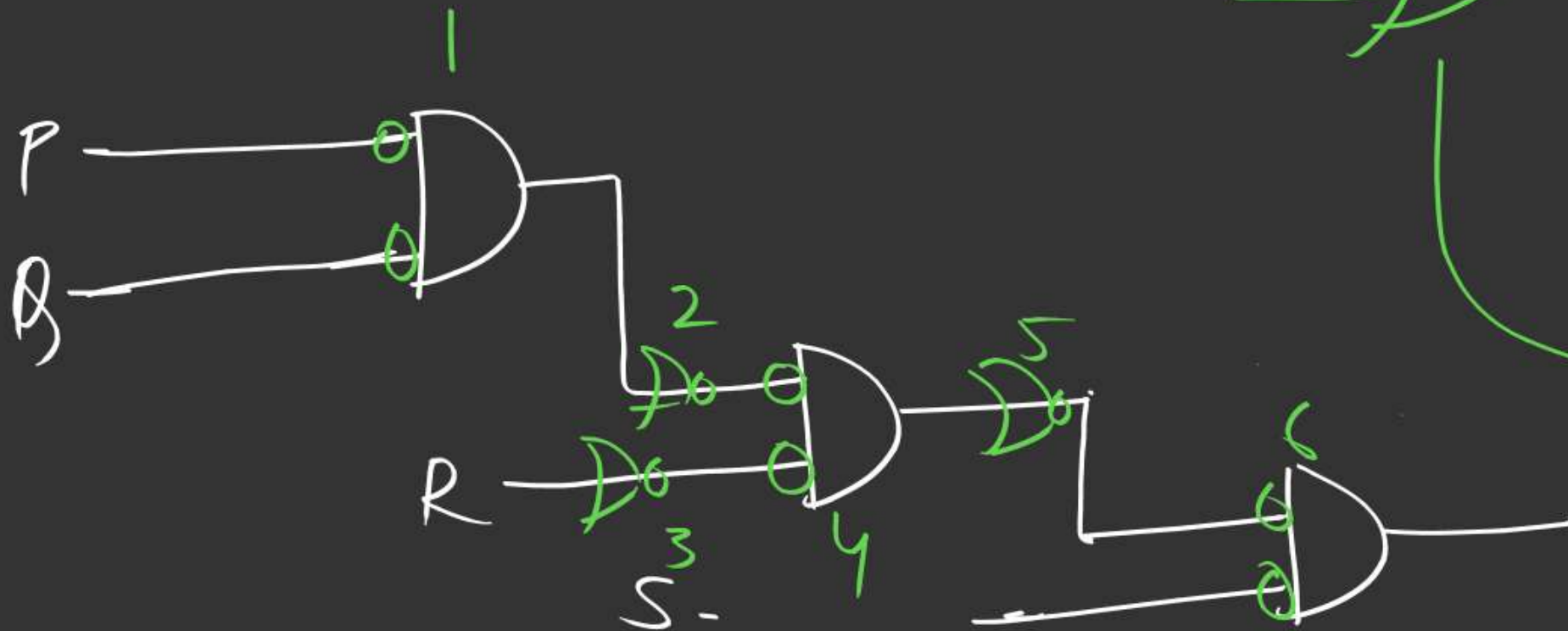
$$y = \overline{P} \cdot \overline{Q} \cdot R \cdot \overline{S}$$

NOR

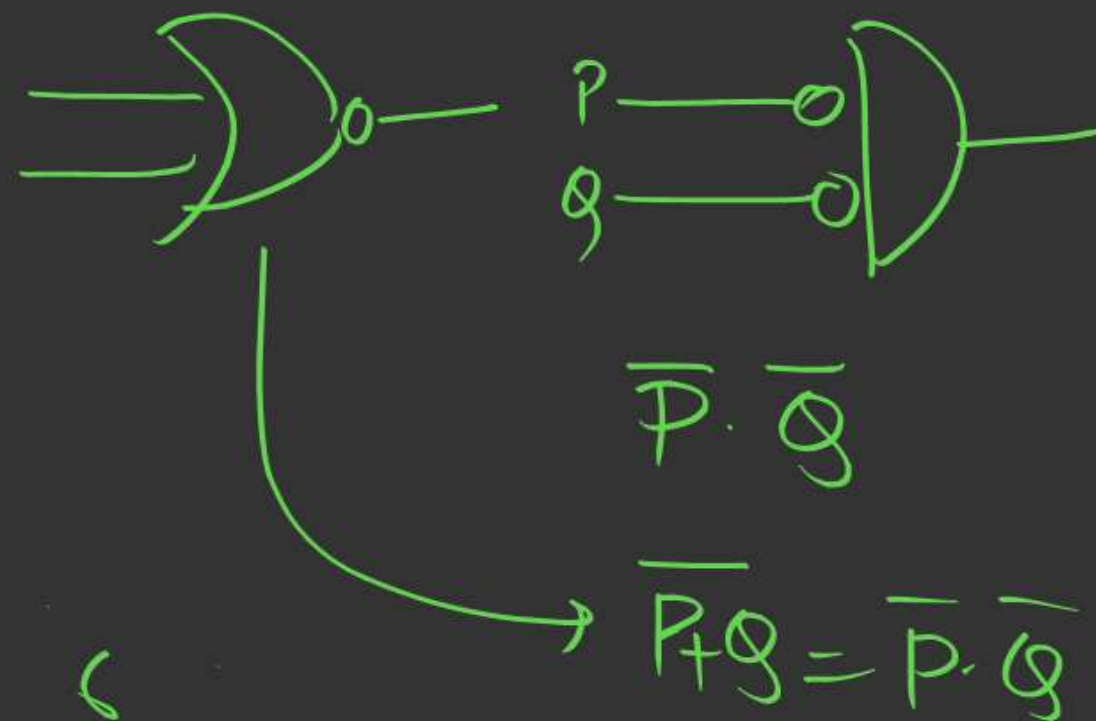
NOT $\rightarrow$ 1

AND $\rightarrow$ 3

OR $\rightarrow$ 2



6 NOR



Case 2:- all i/p's are in OR format.

eg:- $y = P + Q + \bar{R} + S \dots$

1) Min no. of NOR gates : $(2n-2)+K$.

2) Min no. of NAND gates : $(3n-3)-K$.

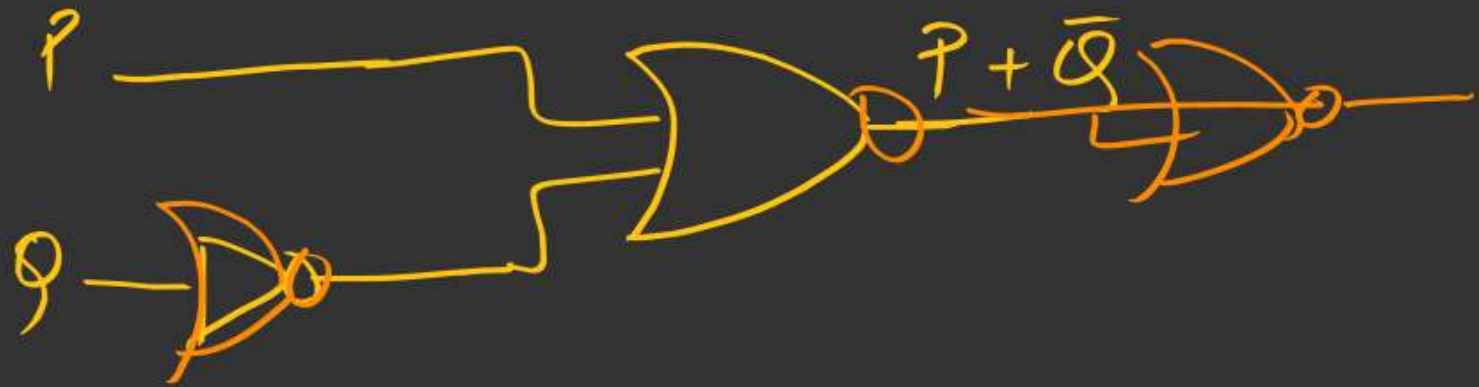
n = Total i/p variable

K = no. of Compl variable.

eg 1: $y = P + \bar{Q}$

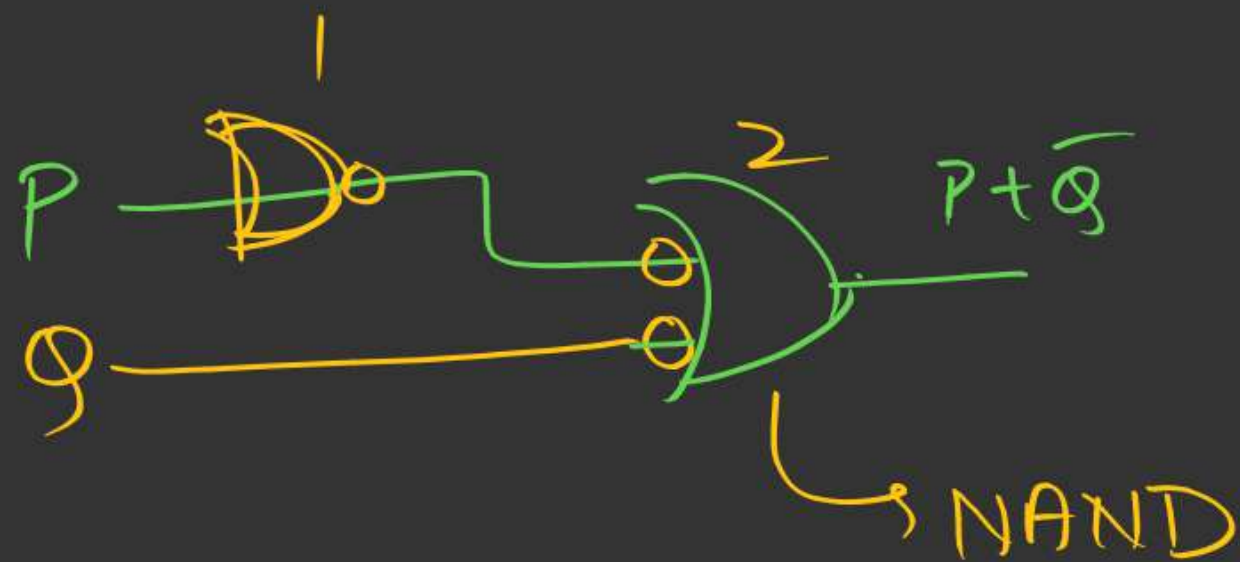
$$n=2, k=1$$

$$\begin{aligned} \text{NOR} &= (2n-2) + k \\ &= (2 \times 2 - 2) + 1 \\ &= 4 - 2 + 1 = \textcircled{3} \end{aligned}$$

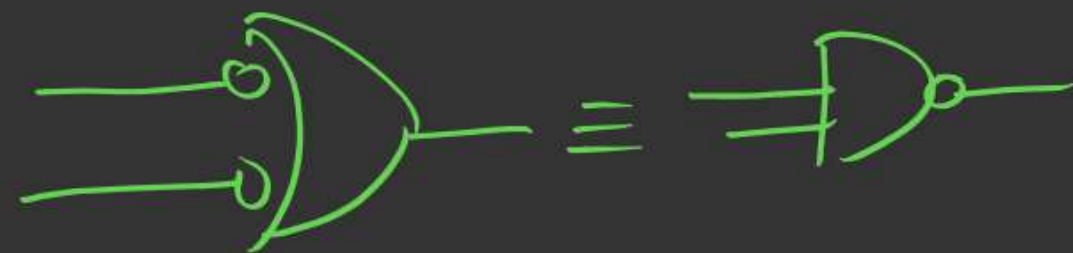


1 NAND

$$y = P + \overline{Q}$$



2 NAND



$$n=2, k=1$$

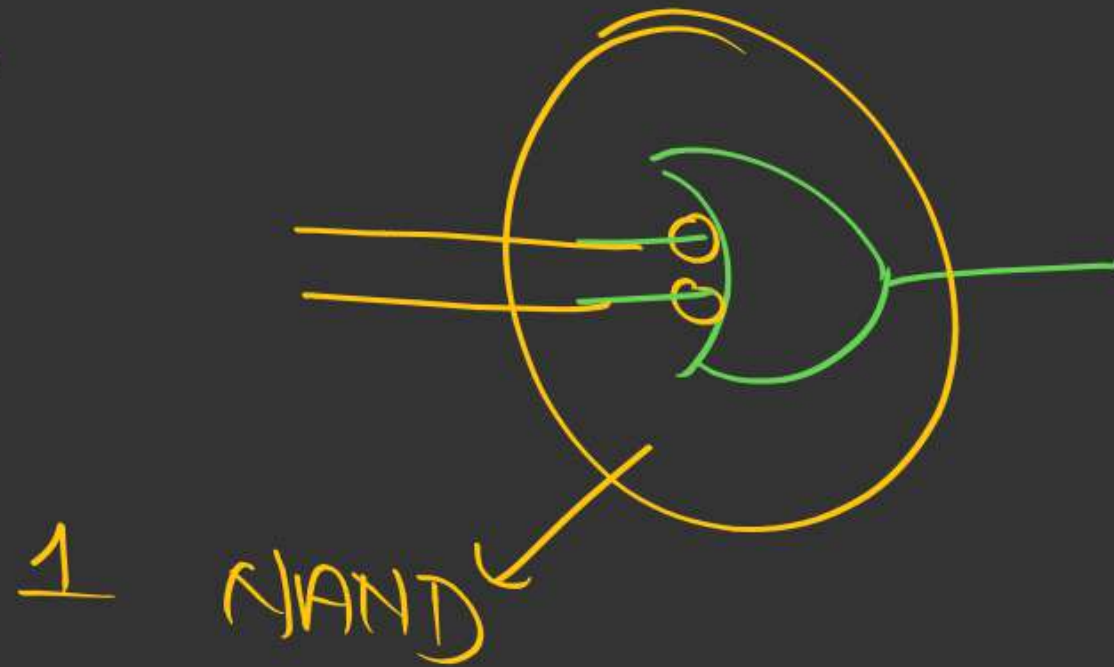
$$(3n-3)-k$$

$$= 3 \times 2 - 3 - 1$$

$$= 6 - 4 = \textcircled{2}$$

$$\text{eq 2)} \quad y = \overline{P} + \overline{Q}$$

NIAND



$$n=2, k=2$$

$$(3n-3) - k$$

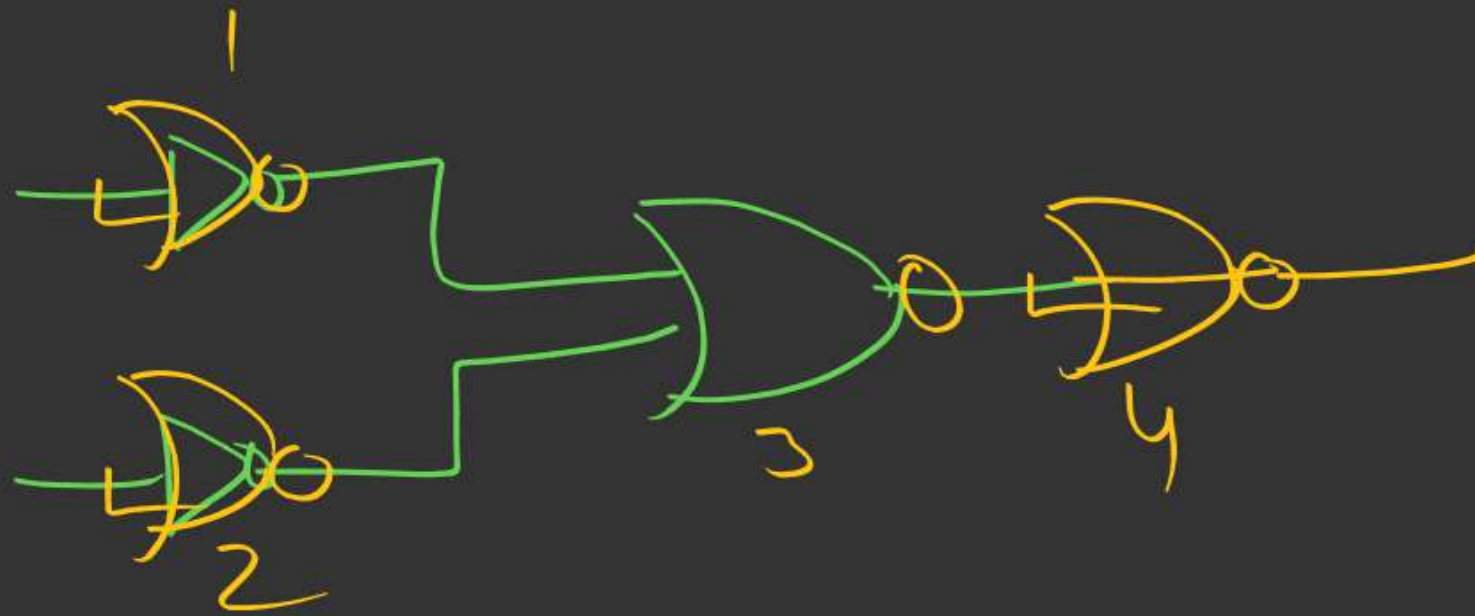
$$3 \times 2 - 3 - 2$$

$$6 - 3 - 2$$

$$= 1$$

NOR

$$y = \overline{P + Q} = \overline{P \cdot Q} = \text{NAND}$$



$\Rightarrow$ 4 NOR

$$y = \bar{P} + Q + \bar{R}$$

NAND

= 4 NAND

$$n = 3$$

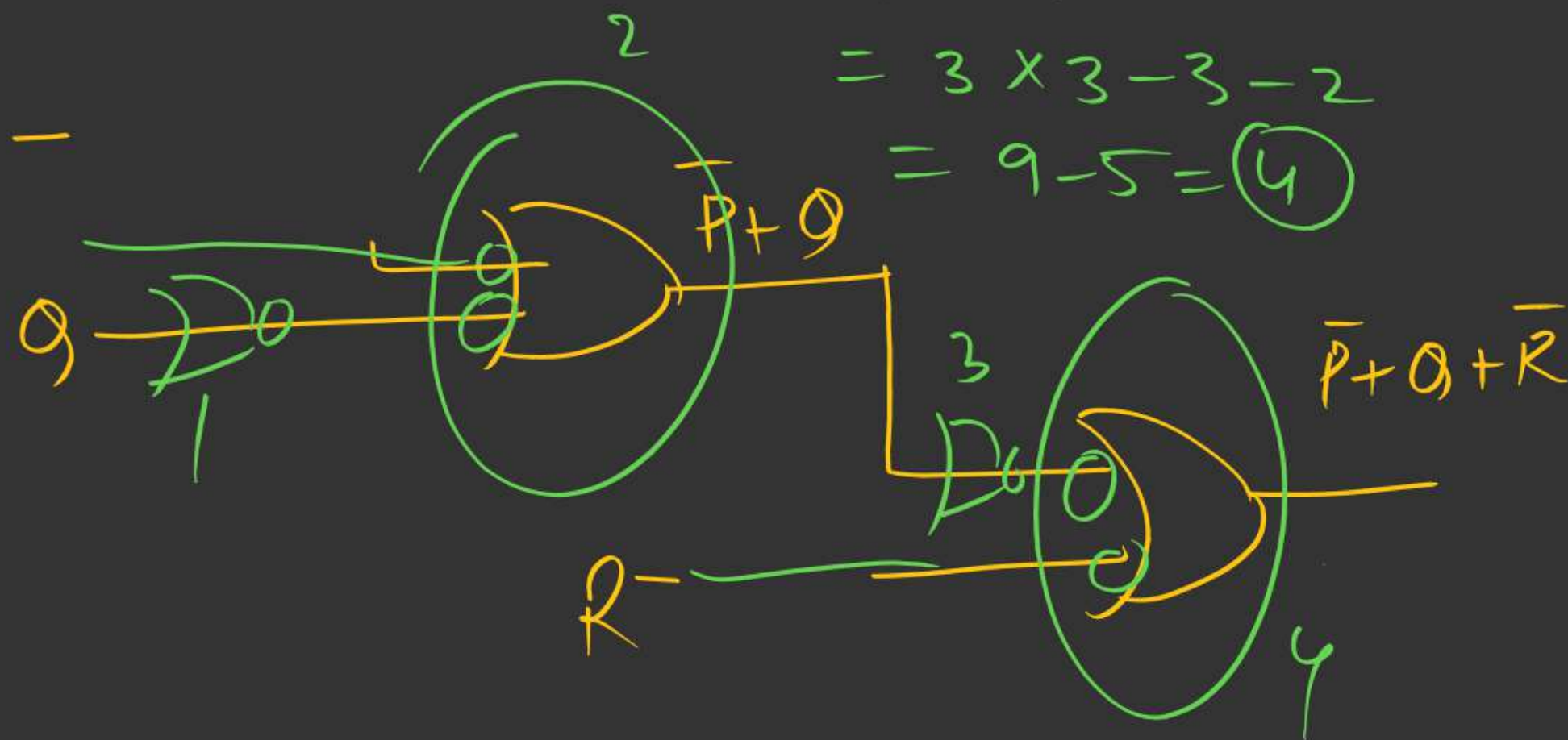
$$k = 2$$

$$(3n - 3) - k$$

$$= 3 \times 3 - 3 - 2$$

$$= 9 - 5 = 4$$

85%



$$y = \bar{P} + Q + \bar{R}$$

100%

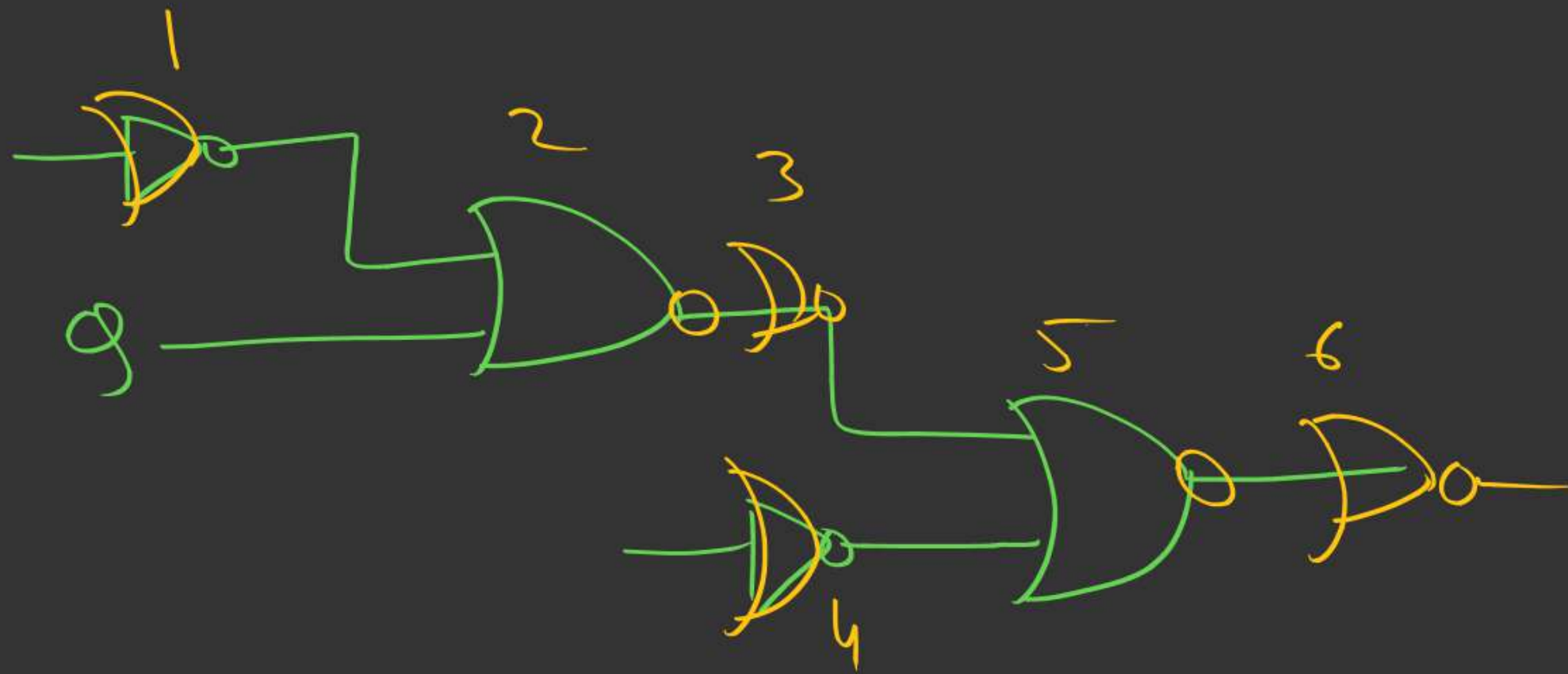
min NOR

6

$$n=3, k=2$$

$$(2n-2)+k$$

$$2 \times 3 - 2 + 2 = 6$$



HW

$$y = P + QR$$

min NAND

Tomorrow →



2 mins Summary



min. no. of NAND & NOR.



THANK - YOU

